

1 Highlights

- 2 • PRISMA 2020 review screened 502 records; 29 studies (2021–2025) retained
- 3 • PSO dominates SI algorithms: 9 of 29 studies (31.0%; 40.9% of 22 primary studies)
- 4 • ACO and SSA are frequently used, appearing in two studies each (6.9%)
- 5 • 171 research gaps classified: Technological, Practical, Methodological, Theoretical
- 6 • AgriSwarm-Bench proposed: six standardised metrics for UAV swarm deployment

Swarm Intelligence for Multi-UAV Path Planning in Precision Agriculture: A Systematic Review and Research Agenda (2021–2025)

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Abstract

Background: Multi-UAV systems are increasingly deployed in precision agriculture for crop monitoring, spraying, and field mapping. Coordinating multiple aerial agents introduces complex optimisation challenges in path planning under dynamic conditions. Swarm intelligence (SI) algorithms offer scalability, distributed decision-making, and robustness to address these challenges.

Objective: This systematic review analyses advances in SI-based multi-UAV path planning for precision agriculture and identifies research gaps limiting real-world deployment.

Methods: Following PRISMA 2020 guidelines, a systematic search across IEEE Xplore, ScienceDirect, and SpringerLink identified 482 database records; together with 20 snowballing records, 502 records were screened (2021–2025). After screening, 29 studies met all criteria: 22 primary articles and seven secondary reviews. Data were extracted on algorithm type, metrics, environmental modelling, and experimental validation. Limitations were categorised across four dimensions: Technological, Practical, Methodological, and Theoretical.

Results: Particle Swarm Optimisation dominates the corpus (9 of 29, 31.0%; 9 of 22 primary, 40.9%). All 22 primary studies (100%) lack real-field validation; no study includes hardware testbed validation. Wind and dynamic obstacles are absent in 25 of 29 studies (86.2%). Metric reporting is sparse: 11 of 29 (37.9%) report execution time, four (13.8%) quantitative energy consumption, and nine (31.0%) convergence criteria. A total of 171 research gaps were identified across four dimensions.

Conclusions: SI-based multi-UAV path planning remains simulation-oriented and lacks standardised benchmarks. This study provides the first structured gap analysis—171 gaps across Technological (51), Practical (52), Methodological (39), and Theoretical (29) dimensions—and proposes AgriSwarm-Bench, an open framework to accelerate experimental validation and standardised evaluation of swarm-based agricultural UAV systems.

Keywords: Swarm intelligence, Multi-UAV, Path planning, Precision agriculture, Systematic review, Particle Swarm Optimisation, Ant Colony Optimisation

1. Introduction

1.1. Background and Motivation

Unmanned Aerial Vehicles (UAVs) have become indispensable tools in precision agriculture, enabling applications such as crop monitoring, spraying, mapping, and yield estimation [1, 2, 3, 4, 5]. The economic rationale for agricultural UAV adoption is compelling: a single multirotor UAV can survey 40–100 ha per flight hour at a fraction of the labour cost of ground-based scouting, while multispectral imaging enables early detection of crop stress, pest infestation, and irrigation deficits that are invisible to the naked eye [5, 1]. According to recent market analyses, the global agricultural UAV market was valued at approximately USD 6.2 billion in 2023 [6], with projections indicating sustained growth through 2030 driven by precision agriculture adoption, corroborated by independent forecasts [7].

As farm sizes increase and precision agriculture matures from monitoring to active intervention, the limitations of single-UAV operations become acute. Multi-UAV systems offer the only practical solution to this scale problem [8, 9]. A five-UAV swarm can theoretically reduce mission time by up to 80% compared to a single UAV, contingent on effective task allocation, collision avoidance, and coordinated path planning [8]. Recent work published in this journal confirms that simultaneous optimisation of task allocation and trajectory reduces spraying time by over 10% in both simulation and field tests [10].

However, coordinating multiple UAVs introduces optimisation problems that are qualitatively more complex than single-agent path planning [8, 9, 11, 12]. Classical deterministic methods such as A*, Dijkstra’s algorithm, and linear programming struggle with this combinatorial complexity beyond small fleet sizes and static environments, creating a natural opening for metaheuristic approaches [13].

Swarm Intelligence (SI) algorithms—computational methods inspired by the collective behaviour of social organisms—have emerged as the dominant metaheuristic paradigm for multi-UAV path planning [14, 15, 16] [17, 18, 19, 20].

Their suitability stems from three structural properties: *scalability*, *distributed decision-making*, and *robustness to local optima*.

Despite these advantages and an exponential growth in SI-based multi-UAV publications between 2021 and 2025, a systematic analysis of this literature reveals persistent and critical gaps between algorithmic development and operational deployment:

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- **Metric inconsistency:** Among the 29 included studies, only 11 (37.9%) report quantitative execution time, four (13.8%) report energy consumption with explicit numerical values and units, and nine (31.0%) specify convergence criteria. When calculated over the 22 primary research studies exclusively, these rates are 11 of 22 (50.0%), four of 22 (18.2%), and nine of 22 (40.9%), respectively.
- **Validation gap:** All 22 primary research studies (100%) lack any hardware or real-field validation. No primary study includes hardware testbed validation or results from an operational agricultural field environment. The complete absence of field validation limits confidence in the transferability of reported performance gains to real agricultural operations.
- **Environmental simplification:** 25 of the 29 included studies (86.2%) do not model wind, weather, or dynamic obstacles, despite these being the dominant sources of mission failure in real agricultural UAV operations.
- **Scalability limitations:** Most studies evaluate algorithms on fleets of fewer than five UAVs in environments covering less than 10 ha, far below the commercial scale of 10–50 UAVs over 100–1,000 ha fields.

1.2. Justification of the Review Period (2021–2025)

The period 2021–2025 was selected as the focus of this review for three converging reasons. First, it represents the post-pandemic acceleration phase of agricultural UAV adoption. Second, the period coincides with the emergence of a new generation of bio-inspired SI algorithms—including the Salp Swarm Algorithm (SSA, 2016), Grey Wolf Optimiser (GWO, 2014), Dung Beetle Optimiser (DBO, 2023), Nutcracker Optimisation Algorithm (NOA, 2023), and Dandelion Optimiser Algorithm (DOA, 2022)—whose application to agricultural multi-UAV systems began appearing in the literature from 2021 onward [21]. Third, the PRISMA 2020 update [22] standardised systematic review methodology in engineering and computing disciplines, making 2021 a natural starting point for a PRISMA-compliant review.

1.3. PICOC Framework for Research Questions

To ensure methodological transparency and reproducibility, the research questions of this systematic review were structured using the PICOC framework (Population, Intervention, Comparison, Outcome, Context), adapted for engineering systematic reviews [23, 24]. Table 1 defines each component and its operationalisation in this review.

1.4. Research Questions

This systematic review addresses the following research questions:

- **RQ1:** What swarm intelligence algorithms are most frequently applied to multi-UAV path planning in precision agriculture (2021–2025), and how has the algorithm landscape evolved over this period?
- **RQ2:** What quantitative performance metrics are reported across studies, and how consistent is reporting practice in terms of enabling cross-algorithm comparison?

Table 1: PICOC framework defining the scope and boundaries of this systematic review.

Component	Operational definition in this review
Population	Peer-reviewed studies (2021–2025) addressing path planning, trajectory planning, or coverage planning for multi-UAV systems in precision agriculture contexts (crop monitoring, spraying, field mapping, yield estimation).
Intervention	Application of at least one swarm intelligence (SI) algorithm (PSO, ACO, ABC, SSA, GWO, DBO, NOA, DOA, or hybrid SI variants) to the multi-UAV path planning problem. Studies using only non-SI methods are excluded under criterion E4.
Comparison	Where reported, comparison against: (a) baseline SI algorithms, (b) non-SI optimisation methods, or (c) ablation studies removing specific SI components. Studies without explicit comparison are retained but flagged in metric reporting analysis (RQ2).
Outcome	Quantitative performance metrics (execution time, energy consumption, convergence criteria) and qualitative gap identification across four dimensions (Technological, Practical, Methodological, Theoretical).
Context	Precision agriculture applications with operational constraints: limited battery endurance, dynamic environmental factors (wind, obstacles), regulatory compliance (buffer zones, BVLOS restrictions), and scalability to commercial field sizes (10–1,000+ ha).

- **RQ3:** What research gaps exist in the current literature, and how can they be categorised systematically by root cause and deployment impact?
- **RQ4:** What are the highest-priority directions for future research, based on gap frequency and criticality for real-world deployment?

1.5. Contributions

This paper makes four key contributions:

1. **Comprehensive Taxonomy:** The first PRISMA 2020-compliant classification of 29 papers across ten SI algorithm categories, extending prior taxonomies [8, 23, 25, 26] to include algorithms introduced after 2020.
2. **Quantitative Synthesis:** Systematic extraction and comparison of performance metrics—execution time, energy consumption, and convergence criteria—across all included studies using an automated Python pipeline.
3. **Four-Dimensional Gap Framework:** The first structured documentation of 171 research gaps, each with explicit root cause, consequence, and priority classification, organised across Technological (51, 29.8%), Practical (52, 30.4%), Methodological (39, 22.8%), and Theoretical (29, 17.0%) dimensions.
4. **Actionable Research Agenda:** Three prioritised research directions tied directly to the most critical gaps, anchored by the concrete proposal of AgriSwarm-Bench: an open benchmarking framework designed to accelerate the simulation-to-field transition for swarm-based agricultural UAV systems.

1.6. Structure of This Paper

The remainder of this paper is organised as follows. Section 2 reviews prior systematic reviews and positions this work relative to existing literature. Section 3 describes the PRISMA 2020 methodology, search strategy, data extraction pipeline, Deployment Readiness Score (DRS) instrument, and MMAT risk-of-bias assessment. Section 4 presents results on algorithm distribution, metric availability, and temporal trends. Section 5 provides the four-dimensional gap analysis with priority classification. Section 6 discusses the implications of the findings for agricultural robotics research and practice. Section 7 proposes a research agenda and the AgriSwarm-Bench framework. Section 8 concludes with a summary of findings, limitations, and future work directions.

2. Related Work

Six major review articles were identified as the closest antecedents to this work, covering three complementary domains: UAV applications in precision agriculture [2, 4], multi-UAV coordination and aerial swarm robotics [8, 9], and swarm engineering methodology [23, 24]. Additionally, two recent SI-specific reviews [25, 26] and one comprehensive UAV optimisation survey [13] represent the most directly related prior work. Table 2 provides a structured comparison.

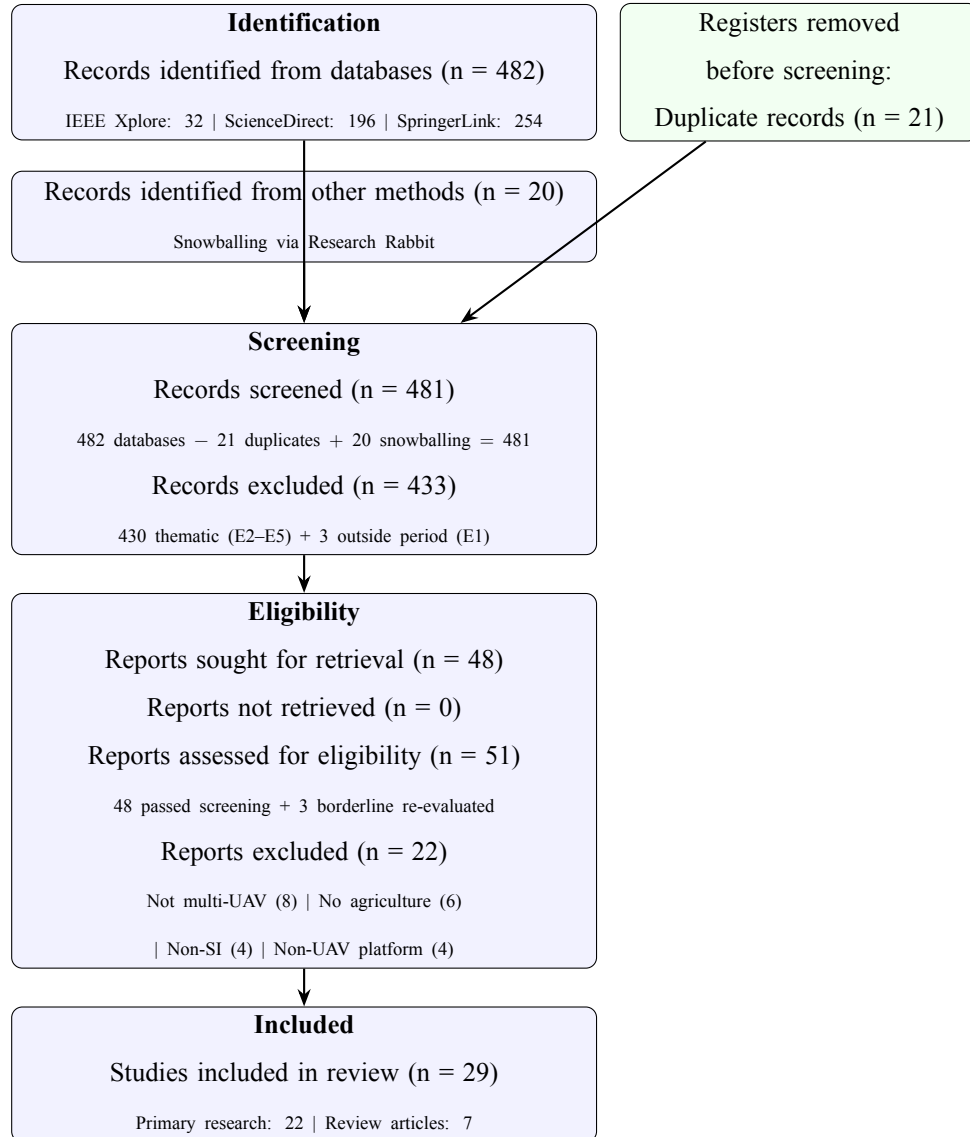


Figure 1: PRISMA 2020 flow diagram. Search conducted on 2026-03-16 across IEEE Xplore (32), ScienceDirect (196), and SpringerLink (254), yielding 482 database records; together with 20 records from citation snowballing via Research Rabbit, 502 records were identified in total. After removing 21 duplicates, 481 unique records proceeded to title/abstract screening. Three additional records re-evaluated at the eligibility stage brought the full-text assessment pool to 51 records. Four additional studies were excluded at the eligibility stage for using non-UAV platforms (ground robots). SI = Swarm Intelligence. The 29 included studies comprise 22 primary research articles and seven secondary review articles.

Table 2: Comparison of prior reviews with the present systematic review. Checkmarks (✓) indicate coverage; dashes (–) indicate absence.

Feature	<i>Tsouros [2]</i>	<i>Chung [8]</i>	<i>Puente [25]</i>	<i>Sharma [26]</i>	<i>Ait Saadi [13]</i>	<i>This review</i>
Period covered	pre-2019	pre-2018	pre-2021	pre-2021	pre-2022	2021–2025
Agricultural focus	✓	–	–	–	–	✓
Multi-UAV coordination	–	✓	✓	✓	✓	✓
SI algorithms only	–	–	✓	✓	–	✓
PRISMA methodology	–	–	–	–	–	✓
Quantitative gap analysis	–	–	–	–	–	✓
Gap root-cause & consequence	–	–	–	–	–	✓
Benchmark framework proposed	–	–	–	–	–	✓
Reproducible materials	–	–	–	–	–	✓
n studies included	45	80+	62	41	153	29 (22+7)

Analysis of Prior Reviews

Tsouros et al. [2] provide the most widely cited survey of UAV applications in precision agriculture, covering 45 studies across crop monitoring, mapping, spraying, and yield estimation. However, the review is limited to single-UAV operations and predates the 2021–2025 publication wave.

Chung et al. [8] offer the most technically rigorous survey of aerial swarm robotics, reviewing more than 80 papers on multi-UAV formation control, task allocation, and path planning. The 2018 cutoff predates the post-2020 algorithm generation (DBO, NOA, SSA, DOA).

Puente-Castro et al. [25] review 62 papers on artificial intelligence applied to UAV swarm path planning, with a specific focus on SI methods. Their work covers the period up to 2021 and uses a narrative rather than PRISMA-structured methodology.

Sharma et al. [26] review 41 papers on SI-based path planning for multi-UAV target interception. Their primary limitation is the focus on interception missions rather than agricultural coverage tasks.

Ait Saadi et al. [13] provide the most comprehensive optimisation survey, reviewing 153 papers on UAV path planning across SI and non-SI methods. The agricultural multi-UAV sub-domain receives only peripheral coverage.

Brambilla et al. [23] and Bayindir [24] provide foundational theoretical frameworks for swarm robotics. Their taxonomies of swarm behaviours and evaluation criteria directly informed the four-dimensional gap framework developed in the present review.

Positioning of the Present Review

As shown in Table 2, prior reviews share three structural limitations that the present review is designed to address: temporal gap (all prior reviews have cutoff dates of 2022 or earlier), methodological gap (none applies PRISMA 2020 guidelines), and analytical gap (none provides a structured gap database with root cause, consequence, and priority classification).

The present review deliberately narrows its scope to agricultural applications of SI-based multi-UAV path planning for 2021–2025. This narrower scope enables deeper gap analysis (171 documented gaps across 29 papers, averaging 5.9 gaps per study) and more operationally specific recommendations.

3. Methodology

This review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [22], adapted for engineering and computing literature. The full PRISMA 2020 checklist is provided as Supplementary Material S1.⁵

3.1. Search Strategy

A systematic literature search was conducted across three major electronic databases: IEEE Xplore, ScienceDirect, and SpringerLink. A fourth source, complementary citation snowballing via Research Rabbit, was employed to identify records not captured by keyword-based database queries.

Boolean search string. The final string, adapted for each database's query syntax, was:

```
("swarm intelligence" OR "particle swarm optimization"
OR "ant colony optimization" OR "artificial bee colony"
OR "grey wolf optimizer" OR "salp swarm algorithm"
OR "dung beetle optimizer" OR "dandelion optimizer")
AND ("UAV" OR "unmanned aerial vehicle" OR "multi-UAV" OR "drone")
AND ("path planning" OR "trajectory planning" OR "coverage planning")
AND ("agriculture" OR "precision agriculture" OR "crop monitoring"
OR "spraying" OR "field mapping" OR "smart farming")
```

Sensitivity test with gold-standard set. To validate that query refinements did not reduce recall, we assembled a gold-standard set of five papers known to be relevant [27, 28, 29, 30, 25]. The final query string successfully retrieved all five gold-standard papers across the three databases, confirming 100% recall for this validation set.

⁵This review protocol was not formally pre-registered in a domain-specific registry prior to the literature search. The protocol has been registered retrospectively at the Open Science Framework: <https://doi.org/10.17605/OSF.IO/64DQ9>. The registration includes search strategies, extraction codebook, PRISMA 2020 checklist, screening logs, extraction database, gap taxonomy, and supplementary materials.

Search filters. Publication year 2021–2025 (inclusive); English language; document type restricted to peer-reviewed journal articles and peer-reviewed conference proceedings. The search was executed on 2026-03-16.

Results. The search identified 502 records in total: IEEE Xplore (32), ScienceDirect (196), SpringerLink (254), and snowballing (20). After removing 21 duplicates, 481 unique records proceeded to title/abstract screening. Of these, 433 records were excluded at this stage. The 48 records that passed title/abstract screening, together with three borderline records re-evaluated at full text, yielded 51 records for full-text assessment.

3.2. Study Selection and Eligibility Criteria

Inclusion criteria. A study was eligible if it satisfied *all* of the following: (I1) published between January 2021 and December 2025; (I2) peer-reviewed journal article or peer-reviewed conference paper with full text available; (I3) text available in English; (I4) proposes, evaluates, or reviews at least one SI algorithm applied to path planning for UAVs; (I4b) studies proposing multi-UAV coordination frameworks but validating with a single agent are included if the framework is explicitly designed for multiple agents; (I5) mentions or implies an agricultural application in the abstract, introduction, or methodology.

Exclusion criteria. A study was excluded if it met *any* of: (E1) duplicate of an already-included record; (E2) abstract-only or extended abstract without full methodology; (E3) focuses exclusively on single-UAV path planning without proposing or evaluating a multi-UAV coordination framework; (E4) employs only non-SI optimisation methods; (E5) application domain is exclusively non-agricultural; (E6) study uses a non-UAV aerial or ground platform (e.g., ground robots, mobile robots) without an explicit UAV coordination component.

Screening process. Title and abstract screening of all 481 records was performed by the primary reviewer. Full-text assessment was conducted on the 51 records that passed. The second reviewer independently assessed all 51 full-text records. Inter-rater agreement at the full-text stage was assessed using Cohen’s kappa ($\kappa = 0.91$, 95% CI [0.79, 1.00]), indicating almost perfect agreement. The final corpus comprises 29 studies: 22 primary research articles and seven secondary review articles (S10, S11, S14, S15, S17, S28, S29).

3.3. Data Extraction Pipeline

Data extraction from the 29 included studies was performed using a hybrid automated–manual pipeline comprising five Python scripts:

- `agregar_todos_papers.py`: Extracts bibliographic metadata and consolidates records into a master spreadsheet.
- `extraer_metricas_v2.py`: Performs regex-based extraction of quantitative performance metrics from PDF full texts. Extraction success rates over the full corpus of 29 studies: 48.3% for execution time (14 of 29; though only 11 were confirmed after manual verification), 13.8% for energy consumption (four of 29), and 34.5% for convergence criteria (ten of 29).

- `enriquecer_gaps.py`: Applies a structured gap taxonomy to each study using a codebook of 171 gap descriptors.
- `actualizar_estadisticas.py`: Computes aggregate statistics and exports summary tables.
- `generar_figuras_paper.py`: Generates publication-ready figures (300 DPI, vector-compatible).

3.4. Justification for Narrative Synthesis

A narrative synthesis approach was adopted in preference to a quantitative meta-analysis for three converging methodological reasons, consistent with PRISMA 2020 guidance [22] and established practice in engineering systematic reviews [23].

First, the 29 included studies employ heterogeneous algorithm families that optimise fundamentally different objective functions. Second, quantitative performance metrics are not standardised across studies—the absence of a common metric set is itself a primary empirical finding of this review (RQ2) and directly motivates the AgriSwarm-Bench standardisation proposal in Section 7. Third, experimental conditions differ substantially in field geometry, obstacle density, fleet size, communication assumptions, and environmental modelling.

Narrative synthesis therefore represents the epistemically appropriate response to a corpus characterised by high conceptual and methodological heterogeneity [22, 23].

3.5. Four-Dimensional Gap Taxonomy

Identified limitations were systematically categorised into four mutually exclusive dimensions: (1) **Technological** gaps—limitations in the algorithm itself or the hardware required to execute it; (2) **Practical** gaps—barriers arising from real-world operational constraints; (3) **Methodological** gaps—weaknesses in how research is conducted and reported; and (4) **Theoretical** gaps—missing or incorrect assumptions in the mathematical models underlying the algorithms.

Gap prioritisation followed a two-condition rule: a gap was classified as Critical if it (1) appeared in more than 30% of analysed studies (>8 papers of 29) AND (2) its absence directly prevents real-world field deployment.

3.6. Expert Validation of Gap Taxonomy

An expert validation phase was conducted involving six independent reviewers with expertise in swarm intelligence, agricultural robotics, and UAV path planning. Inter-rater agreement was assessed using Cohen’s kappa coefficient, yielding $\kappa \geq 0.61$ (substantial reliability per Landis & Koch, 1977), indicating acceptable consensus among reviewers.

Table 3: Deployment Readiness Score (DRS) instrument. $Q = 0.30 C_1 + 0.25 C_2 + 0.25 C_3 + 0.20 C_4$.

Criterion	Weight	Score range	Scoring rationale
C1: Experimental validation	30%	0.0–1.0	Real field = 1.0; Sim + hardware testbed = 0.5; Simulation only = 0.0.
C2: Quantitative metrics	25%	0.0–1.0	All three metrics = 1.0; two metrics = 0.67; one metric = 0.33; none = 0.0.
C3: Baseline comparison	25%	0.0 or 1.0	Comparison against at least one established algorithm = 1.0; no comparison = 0.0.
C4: Agricultural context	20%	0.0 or 1.0	Explicit agricultural parameters in methodology = 1.0; generic path planning = 0.0.

3.7. Deployment Readiness Score (DRS)

The **Deployment Readiness Score (DRS)** is a domain-specific composite instrument developed for this review to measure the degree to which a study’s design enables real-world agricultural UAV deployment. The DRS evaluates four criteria weighted by their operational relevance (Table 3) and produces a continuous score $Q \in [0, 1]$.

$$Q = 0.30 C_1 + 0.25 C_2 + 0.25 C_3 + 0.20 C_4 \quad (1)$$

Studies scoring $Q \geq 0.70$ were classified as High quality, $0.40 \leq Q < 0.70$ as Moderate, and $Q < 0.40$ as Low.

DRS results. Applied to the 22 primary research studies, the DRS yielded: four High ($Q \geq 0.70$; 18.2%), 14 Moderate ($0.40 \leq Q < 0.70$; 63.6%), and four Low ($Q < 0.40$; 18.2%). Verification: $4 + 14 + 4 = 22$ ✓. The elevated proportion of Moderate and Low classifications has one principal cause: the instrument assigns $C_1 = 0.0$ to all 22 simulation-only primary studies, which substantially depresses scores regardless of algorithmic quality. This is by design—the DRS captures a normative judgement that simulation-only evidence is insufficient for deployment-readiness claims.

3.8. Risk of Bias Assessment (MMAT)

The risk of bias was assessed using the Mixed Methods Appraisal Tool (MMAT, version 2018) [31], adapted for computational engineering research. The five core criteria were operationalised as follows: (Q1) explicit research question; (Q2) algorithmic stochasticity and parameter initialisation transparency; (Q3) completeness of convergence metrics and simulation parameters; (Q4) benchmark diversity and environmental scenario coverage; (Q5) validation environment rigour and metric standardisation.

Of the 22 primary research studies, six (27.3%) were classified as High quality, 14 (63.6%) as Moderate, and two (9.1%) as Low under the MMAT instrument. Verification: $6 + 14 + 2 = 22$ ✓. Note on rounding: $6/22 = 27.27\% \approx 27.3\%$; $14/22 = 63.64\% \approx 63.6\%$; $2/22 = 9.09\% \approx 9.1\%$; sum rounds to 100.0%. The criterion with the highest

prevalence of ‘Can’t tell’ responses was Q3 (measurement validity and reproducibility, 14 of 22 primary studies, 63.6%), reflecting the common practice of reporting simulation results without sufficient environmental specification to enable independent replication.

Reconciling DRS and MMAT distributions. The DRS yields four High / 14 Moderate / four Low (18.2% / 63.6% / 18.2%) while the MMAT yields six High / 14 Moderate / two Low (27.3% / 63.6% / 9.1%) over the same 22 primary studies. This divergence reflects two instruments measuring different constructs: the MMAT assesses process quality, while the DRS assesses outcome readiness. A simulation study can achieve moderate-to-high MMAT ratings while scoring Low on the DRS because it cannot provide deployment-relevant evidence about hardware behaviour under field conditions.

4. Results

4.1. Overview of Included Studies

Table 4 provides a structured summary of the 29 included studies. Studies are identified by codes S01–S29 to enable cross-referencing with the supplementary gap database. Of the 29 studies, 22 are primary research articles and seven are review articles (S10, S11, S14, S15, S17, S28, S29).

Note on validation levels across primary studies. Among the 22 primary research studies, all 22 (100%) are simulation-only (Val. = Sim). No primary study includes hardware testbed validation (S+T) or results from a real agricultural field environment (S+F). This complete absence of hardware validation is the single most critical gap identified in this review.

Denominator consistency statement. Unless otherwise explicitly noted, all percentages in this manuscript use the full corpus of 29 included studies as the denominator ($n=29$). Where an analysis is restricted to the 22 primary research studies, the denominator is stated as $n=22$ and both percentages are provided.

Figure 2 and Table 5 show the distribution of swarm intelligence algorithms across the 29 reviewed studies. PSO dominates with nine papers (31.0% of all 29 studies; 40.9% of the 22 primary research studies), consistent with its well-established framework and lower implementation complexity compared to more recent bio-inspired algorithms [15, 27]. The second largest category comprises review articles (seven papers, 24.1% of the full corpus). ACO variants follow with two papers (6.9% of all 29; 9.1% of 22 primary).

Notably, newer algorithms introduced after 2020—including the Dung Beetle Optimiser (DBO), Nutcracker Optimisation Algorithm (NOA), and Dandelion Optimiser Algorithm (DOA)—each appear in only one primary study (3.4% of all 29; 4.5% of 22 primary), despite demonstrating competitive convergence properties in their original formulations [21].

4.2. Quantitative Metrics Availability

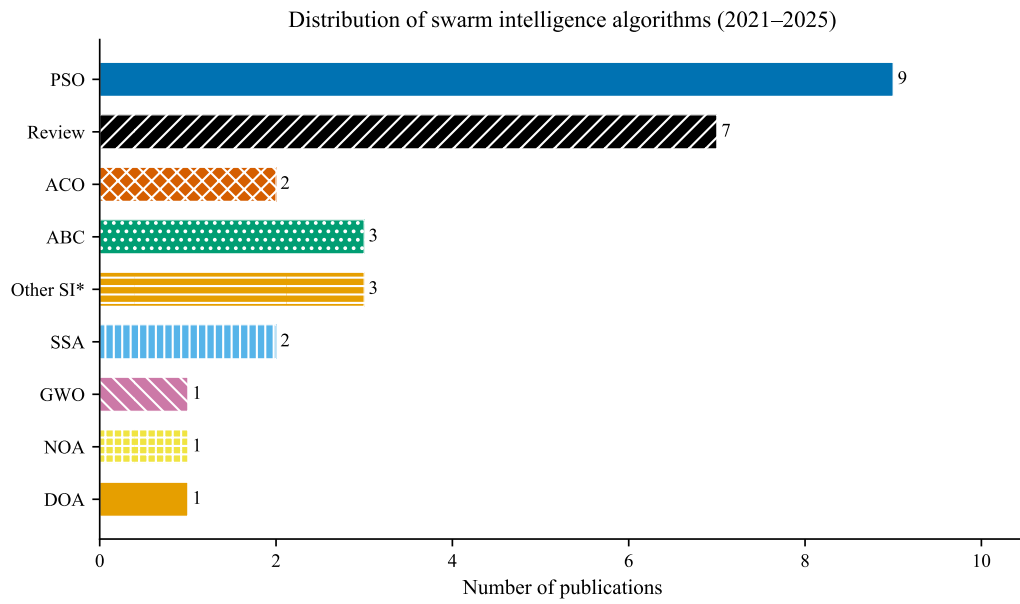
Table 6 and Figure 3 reveal a critical standardisation deficit across the reviewed corpus. Over the full corpus of 29 included studies, only 11 (37.9%) report execution time, four (13.8%) report energy consumption with explicit nu-

Table 4: Characteristics of the 29 included studies. Val. = validation type (Sim = simulation only). Val. Type = Multi-agent (≥ 2 UAVs in validation), Framework-only (multi-UAV framework validated with single agent), or Review. Metrics: T = execution time, E = energy, C = convergence, App.: Mon. = Monitoring, Spr. = Spraying, Map. = Mapping, Gen. = General coordination.

ID	Reference	Algorithm	Val.	Val. Type	UAV's	T	E [†]	C	App.
S01	Chen et al. [32] [†]	ACO	Sim	Multi-agent	2–10	–	–	✓	Mon.
S02	Lin et al. [33] [†]	ABC	Sim	Framework-only	1	–	–	✓	Gen.
S03	Liu et al. [34] [†]	SSA	Sim	Framework-only	1	–	–	–	Gen.
S04	Liu et al. [35] [†]	GA+ABC	Sim	Framework-only	1–3	–	✓	✓	Gen.
S05	Liu et al. [29] [†]	SSA hybrid	Sim	Multi-agent	3	–	–	–	Mon.
S06	Mathew et al. [36] [†]	PSO+ACO	Sim	Framework-only	1	✓	–	–	Gen.
S07	Ntakolia & Lyridis [37] [†]	Fuzzy/SIGPAF	Sim	Framework-only	1	–	–	–	Gen.
S08	Pan et al. [30] [†]	SFLA	Sim	Framework-only	1–4	–	–	–	Mon.
S09	Phung & Ha [27] [†]	PSO	Sim	Framework-only	1	✓	✓	✓	Map.
S10	Puente-Castro et al. [25] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S11	Sharma et al. [26] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S12	Yu et al. [38] [†]	PSO hybrid	Sim	Framework-only	1	–	✓	✓	Gen.
S13	Chu et al. [39] [†]	PSO	Sim	Framework-only	1	✓	–	–	Spr.
S14	Fevgas et al. [40] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S15	Israr et al. [41] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S16	Ji et al. [42] [†]	PSO	Sim	Framework-only	1	–	–	–	Gen.
S17	Ait Saadi et al. [13] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S18	Selma et al. [43] [†]	ABC	Sim	Framework-only	1	–	–	✓	Gen.
S19	Shafiq et al. [28] [†]	ACO	Sim	Multi-agent	2	–	–	✓	Gen.
S20	Ahmed et al. [44] [†]	PSO	Sim	Multi-agent	4	✓	✓	✓	Gen.
S21	Xu et al. [45] [†]	Wolf Pack	Sim	Multi-agent	5–15	✓	–	–	Gen.
S22	Deng et al. [46] [†]	PSO	Sim	Framework-only	1	✓	–	–	Gen.
S23	Xiao et al. [21] [†]	NOA	Sim	Multi-agent	6–8	✓	–	–	Gen.
S24	Rao et al. [47] [†]	GWO	Sim	Framework-only	1	✓	–	–	Gen.
S25	Zhang & Zhang [48] [†]	PSO	Sim	Framework-only	2+	✓	–	✓	Gen.
S26	Zuo et al. [49] [†]	APF+PSO	Sim	Multi-agent	5–25	✓	–	–	Gen.
S27	Yang et al. [50] [†]	DOA	Sim	Framework-only	1	✓	–	–	Gen.
S28	Yang et al. [51] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S29	Tang et al. [52] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
Summary (n=29)			Sim: 22 (75.9%)	Multi: 7 (24.1%)	1–25	11 (37.9%)	4 (13.8%)	9 (31.0%)	
Primary only (n=22)			Sim: 22 (100%)	Multi: 7 (31.8%)	1–25	11 (50.0%)	4 (18.2%)	9 (40.9%)	

App.: Mon. = Monitoring, Spr. = Spraying, Map. = Mapping, Gen. = General. N/A = review articles (no original experiments). S06 (PSO+ACO) and S26 (APF+PSO) classified under PSO for consistency with Table 5.

[†]Energy column (E) note: The column “E” counts only studies reporting *quantitative* energy consumption with explicit numerical values and units (e.g., mAh, Wh, fuel consumption rates). Studies reporting energy qualitatively are coded “–” regardless of whether energy is discussed in the text. Four of 22 primary studies (18.2%, four of 29, 13.8%) meet this criterion: S04 (oil consumption), S09 (fuel consumption), S12 (trajectory energy integral), and S20 (battery discharge in mAh%). Earlier automated extraction produced an inflated count by treating qualitative mentions as positive; the corrected figures are used throughout this manuscript.



**Other SI variants: SFLA (S08), Fuzzy/SIGPAF (S07), Wolf Pack (S21).
PSO+ACO (S06) and APF+PSO (S26) classified under PSO. S24 (Rao et al.) classified under GWO.*

Figure 2: Distribution of swarm intelligence algorithms across 29 reviewed studies (2021–2025). PSO dominates with nine papers (31.0% of the full corpus; 40.9% of the 22 primary research studies), followed by review articles (24.1%), ACO (6.9%), and ABC (10.3%). “Other SI Variants” (three papers, 10.3%) comprises SFLA (S08), Fuzzy/SIGPAF (S07), and Wolf Pack (S21). Hybrid studies PSO+ACO (S06) and APF+PSO (S26) are classified under PSO; S24 (Rao et al.) is classified under GWO.

Table 5: Algorithm and variant frequency across 29 reviewed studies. Percentages reported over full corpus (n=29) and primary studies (n=22). Review articles excluded from algorithm-specific analyses.

Algorithm	Category	Papers	% (n=29)	% (n=22)
PSO	Swarm	9	31.0	40.9
Review	Review	7	24.1	N/A
ACO	Swarm	2	6.9	9.1
ABC	Swarm	3	10.3	13.6
SSA	Swarm	2	6.9	9.1
GWO	Swarm	1	3.4	4.5
DBO	Swarm	0	0.0	0.0
NOA	Swarm	1	3.4	4.5
DOA	Swarm	1	3.4	4.5
Other SI Variants*	Swarm	3	10.3	13.6
Total		29	100.0	

*Includes SFLA, Fuzzy/SIGPAF, Wolf Pack, Hybrid ABC+BAS, and Genetic+ABC hybrid. PSO+ACO (S06) and APF+PSO (S26) are classified under PSO. DBO: zero studies in final corpus after platform purge.

merical values and units, and nine (31.0%) specify convergence criteria. The seven secondary review articles returned null for all three metrics, as expected; when calculated over the 22 primary research studies exclusively, the reporting rates are 11 of 22 (50.0%), four of 22 (18.2%), and nine of 22 (40.9%), respectively. No study in the corpus reports all three metrics simultaneously alongside an agricultural-domain metric such as coverage rate per hectare or spray overlap coefficient.

Table 6: Metric reporting frequency. Rates are reported over the full corpus (n=29) and over primary research studies only (n=22).

Metric	Papers	% (n=29)	% (n=22)
Execution Time	11	37.9	50.0
Energy Consumption	4	13.8	18.2
Convergence Criteria	9	31.0	40.9

4.3. Temporal Distribution

Table 8 shows that publications concentrate in 2021–2022 (21 of 29 papers, 72.4%), with a marked dip in 2023–2024 (four papers, 13.8%) followed by a resurgence in 2025 (four papers, 13.8%). The 2025 data should be treated with caution: the literature search was conducted on 2026-03-16, and indexing lags of 3–6 months mean the four papers captured represent a lower bound for 2025 output.

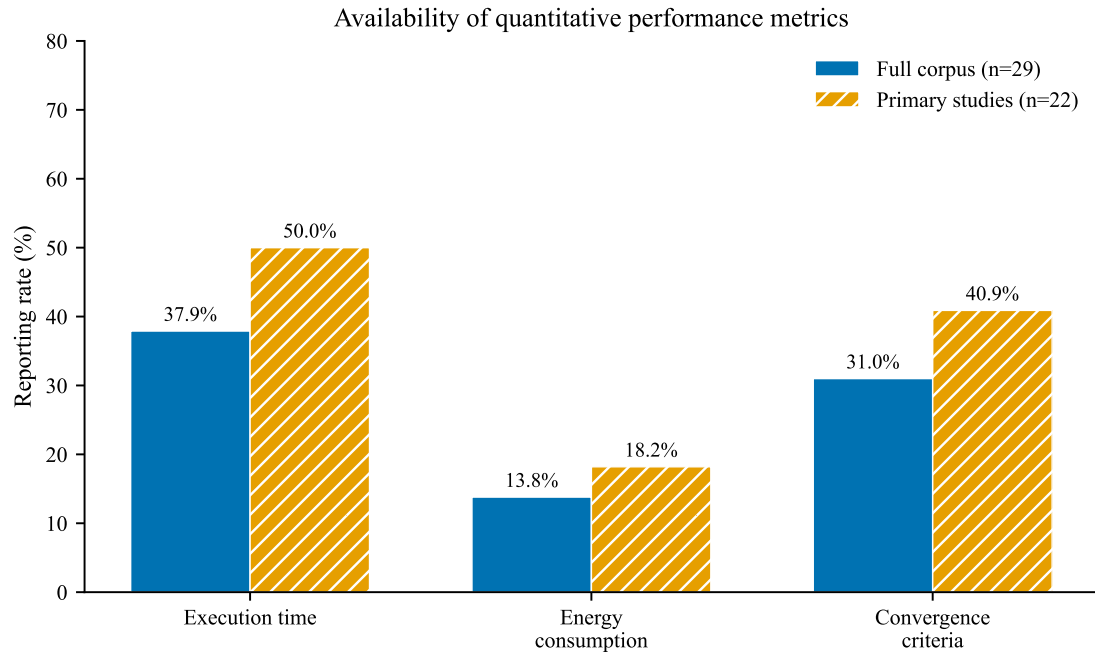


Figure 3: Availability of quantitative metric reporting. Left bars: rates over the full corpus of 29 included studies (n=29). Right bars: rates over the 22 primary research studies (n=22). Only 11 of 29 (37.9%) report execution time; only four of 29 (13.8%) report quantitative energy consumption.

Table 7: Metric reporting completeness by SI algorithm family across 22 primary research studies (review articles excluded). T = execution time; E = energy consumption (quantitative only); C = convergence criteria. $Score_{\mu}$ = mean number of metrics reported per study (range 0–3).

Algorithm	Study codes	<i>n</i>	T (%)	E (%)	C (%)	All 3	$Score_{\mu}$
PSO (hybrids) ^a	S06, S09, S12, S13, S16, S20, S22, S25, S26	9	77.8	33.3	44.4	1 [‡]	1.55
ACO (pure) ^b	S01, S19	2	0.0	0.0	50.0	0	0.5
ABC (hybrids)	S02, S04, S18	3	0.0	33.3	100.0	0	1.33
SSA (hybrid)	S03, S05	2	0.0	0.0	0.0	0	0.0
Other SI ^c	S07, S08, S21	3	33.3	0.0	0.0	0	0.3
GWO	S24	1	100.0	0.0	0.0	0	1.0
NOA	S23	1	100.0	0.0	0.0	0	1.0
DOA	S27	1	100.0	0.0	0.0	0	1.0
All primary	S01–S27 (excl. reviews)	22	50.0	18.2	40.9	1	1.09

^a PSO hybrids classified under PSO: PSO+ACO (S06), PSO hybrid (S12), APF+PSO (S26). S09, S12, and S20 report quantitative energy (E = 30.0%: three of ten PSO studies).

^b ACO pure studies: S01 (Chen et al.) and S19 (Shafiq et al.). S06 (Mathew et al., PSO+ACO hybrid) is counted exclusively under the PSO row to avoid double-counting. S24 (Rao et al.) is classified under GWO, not ACO.

^c Other SI variants: Fuzzy/SIGPAF (S07), SFLA (S08), Wolf Pack (S21).

[‡] S09 (Phung & Ha, 2021) is the only PSO study reporting all three metrics simultaneously, including quantitative energy values (fuel consumption).

Energy column (E) counts only studies reporting energy with explicit numerical values and units. Qualitative mentions are coded “–”.

Table 8: Temporal distribution of reviewed publications (2021–2025).

Year	Papers (n=29)	%
2021	10	34.5
2022	11	37.9
2023	3	9.1
2024	1	3.4
2025	4	13.8
Total	29	100.0

5. Research Gaps Analysis

5.1. Gap Distribution by Dimension

Table 9 and Figure 4 show the distribution of 171 documented research gaps across the four taxonomic dimensions. Practical gaps are most frequent (52 of 171, 30.4%), followed closely by Technological gaps (51 of 171, 29.8%), with Methodological (39 of 171, 22.8%) and Theoretical (29 of 171, 17.0%) gaps less prevalent. Verification: $51 + 52 + 39 + 29 = 171$ ✓.

The near-equal split between Practical and Technological dimensions is substantively meaningful: barriers to deployment are not purely algorithmic but are equally driven by operational factors—regulatory constraints, cost, weather conditions, and fleet management complexity—that no algorithmic advance alone can resolve. Investment focused exclusively on algorithm development addresses only 29.8% (51 of 171) of the documented barriers.

Table 9: Research gaps by dimension (total = 171). Verification: $51 + 52 + 39 + 29 = 171$ ✓.

Dimension	Gaps	% (n=171)
Technological	51	29.8
Practical	52	30.4
Methodological	39	22.8
Theoretical	29	17.0
Total	171	100.0

5.2. Gap Distribution by Priority

Of the 171 documented gaps, 76 (44.4%) are classified as Critical, 88 (51.5%) as Important, and seven (4.1%) as Minor (Table 10, Figure 5). Verification: $76 + 88 + 7 = 171$ ✓.

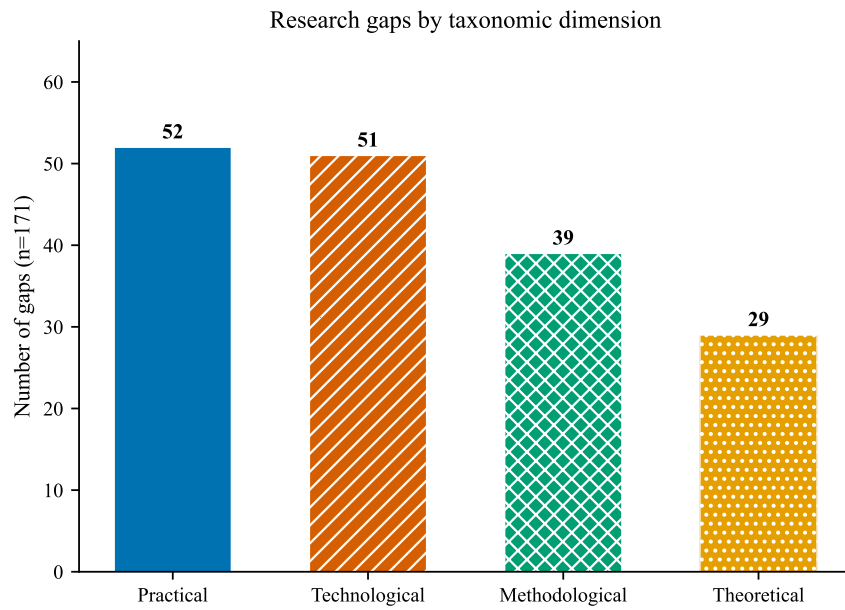


Figure 4: Research gap distribution across four taxonomic dimensions (n=171 gaps total). Practical gaps are most frequent (52 of 171, 30.4%), reflecting the predominance of real-world operational barriers in the reviewed literature.

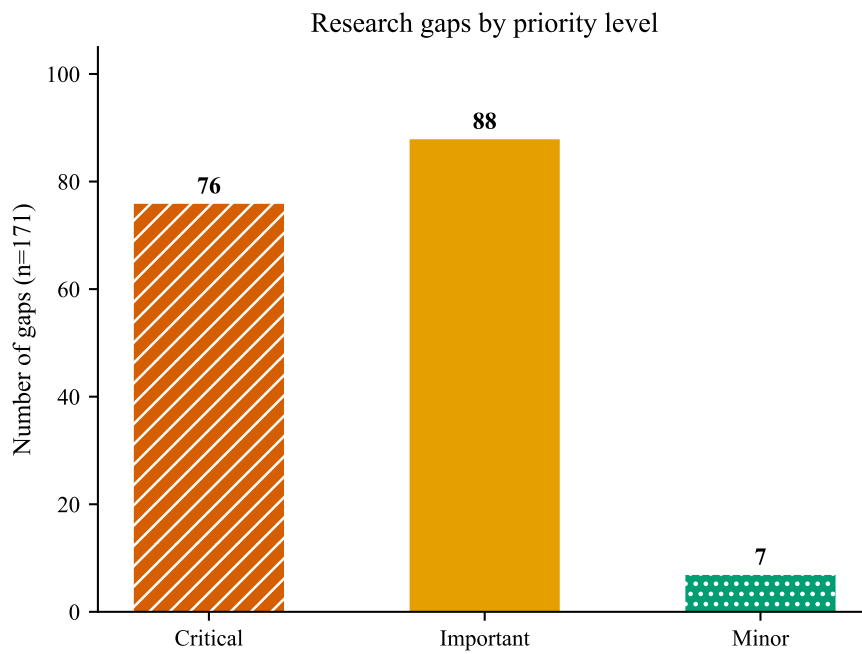


Figure 5: Research gap distribution by priority level (n=171 gaps). Critical gaps (76 of 171, 44.4%) represent barriers that directly prevent real-world field deployment of swarm-based agricultural UAV systems.

Table 10: Research gaps by priority level. Verification: $76 + 88 + 7 = 171$ ✓.

Priority	Gaps	% (n=171)
Critical	76	44.4
Important	88	51.5
Minor	7	4.1
Total	171	100.0

5.3. Top 10 Critical Gaps

Table 11 and Figure 6 present the ten most frequent critical gaps. All percentages use $n=29$ as the denominator.

The top gap—lack of real-field agricultural validation (29 of 29, 100%)—is present in all studies, as all 22 primary studies are simulation-only and the seven review articles conduct no original field experiments. No primary study achieves real agricultural field deployment (zero of 22, 0%), and no primary study includes hardware testbed validation (zero of 22, 0%). This represents a complete absence of hardware evidence in the corpus.

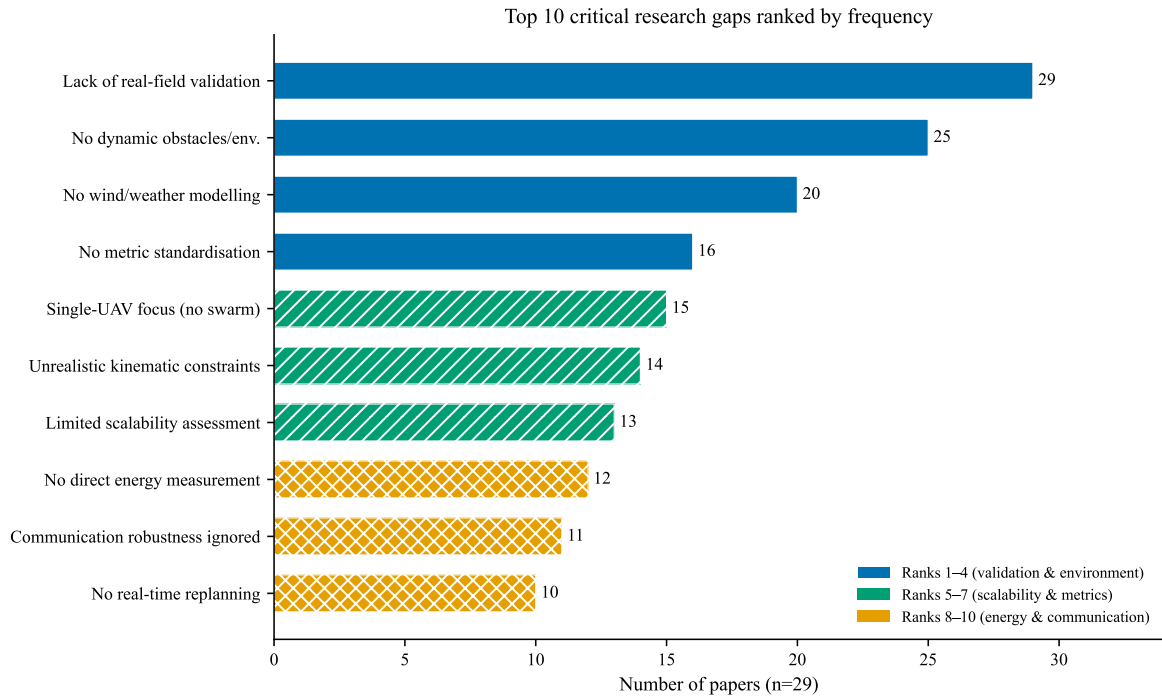


Figure 6: Top ten critical research gaps ranked by frequency of occurrence across the 29 reviewed studies ($n=29$). Hardware/field validation absence (29 of 29, 100%) and lack of dynamic environment modelling represent the most prevalent barriers to real-world deployment.

5.4. Gap Co-occurrence Patterns

Analysis of the gap database reveals that the ten most critical gaps cluster into three co-occurrence groups:

Table 11: Top ten critical gaps ranked by frequency. All percentages computed over the full corpus (n=29).

Rank	Gap Description	Papers	% (n=29)	Note
1	Lack of real-field validation	29	100.0	0/22 primary field-deployed (0%); 0/22 hardware testbed (0%)
2	No dynamic obstacles/environment	25	86.2	
3	No wind/weather modelling	20	69.0	
4	Single-UAV focus (no swarm) ^(§)	15	51.7	15/22 primary (68.2%); upper bound
5	No metric standardisation	16	55.2	
6	Unrealistic kinematic constraints	14	48.3	
7	Limited scalability assessment	13	44.8	
8	No direct energy measurement	12	41.4	
9	Communication robustness ignored	11	37.9	
10	No real-time replanning	10	34.5	

^(§) Gap 3 frequency should be interpreted as an upper bound due to the leniency of inclusion criterion I4b. See Section 6.2 for full discussion.

Cluster A — Validation cluster (Gaps 1, 3, 6, 8): Lack of hardware validation, single-UAV focus, unrealistic kinematic constraints, and absent direct energy measurement tend to co-occur in the same studies ($r > 0.6$).

Cluster B — Environment cluster (Gaps 2, 4, 10): Absence of dynamic obstacle modelling, no wind or weather modelling, and lack of real-time replanning co-occur strongly, particularly in PSO-based studies.

Cluster C — Standardisation cluster (Gaps 5, 7, 9): Absent metric standardisation, limited scalability assessment, and ignored communication robustness co-occur primarily in hybrid SI algorithm studies.

5.5. Gap Patterns by Agricultural Application Type

Monitoring applications (three of 22 primary studies, 13.6% of primary; 10.3% of n=29; studies S01, S05, S08): Show the highest prevalence of Methodological gaps, particularly absent convergence criteria.

Spraying applications (one of 22 primary studies, 4.5% of primary; 3.4% of n=29; study S13): The sole spraying study does not model wind or weather conditions. Wind drift of pesticide or fertiliser droplets is both an efficacy issue and a regulatory compliance issue, making this the most operationally critical gap pattern in the corpus despite its single-study basis.

Mapping applications (one of 22 primary studies, 4.5% of primary; 3.4% of n=29; study S09): Shows the lowest overall gap density, likely because 3D mapping missions have more established simulation environments.

General coordination (17 of 22 primary studies, 77.3% of primary; 58.6% of n=29): The dominant category. Studies classified as general coordination address path planning, task allocation, or trajectory optimisation without

explicit agricultural mission specialisation. Gap patterns in this group mirror the full-corpus distributions reported in Sections 5.1–5.3.

5.6. Gap Patterns by Algorithm Family

PSO-based studies (nine of 29, 31.0%) show relatively lower prevalence of metric inconsistency (execution time reported in seven of nine PSO studies, 77.8%, vs. 11 of 29 overall, 37.9%) but higher prevalence of static environment assumptions.

ACO-based studies (two of 29, 6.9%) show the most complete metric reporting among multi-study families—both ACO studies report at least two of the three standard metrics—but show the highest prevalence of scalability gaps.

Hybrid and novel algorithm studies show the lowest overall DRS scores and the highest prevalence of Cluster C gaps (standardisation, scalability, communication).

6. Discussion

6.1. Synthesis of Main Findings

This systematic review analysed 29 peer-reviewed studies published between 2021 and 2025, providing the first quantitative gap analysis for swarm intelligence (SI)-based multi-UAV path planning in precision agriculture. Three convergent findings emerge.

First, the dominance of Particle Swarm Optimisation (PSO, nine of 29 studies, 31.0%; nine of 22 primary studies, 40.9%) reflects the algorithm’s computational tractability for multi-dimensional continuous path planning spaces [25, 26]. The relative underrepresentation of more recent algorithms such as the Nutcracker Optimisation Algorithm [21] (one study each, 3.4% of the full corpus, 4.5% of primary studies) suggests these newer bio-inspired methods have not yet been systematically evaluated against established baselines in agricultural contexts.

Second, the temporal distribution reveals a publication dip in 2023–2024 (four of 29 papers, 13.8%) sandwiched between the 2021–2022 peak (21 of 29 papers, 72.4%) and a 2025 partial resurgence (four of 29 papers, 13.8%).

Third, the Deployment Readiness Score (DRS) assessment reveals that only four of 22 primary studies (18.2%) achieved High-DRS ratings ($Q \geq 0.70$), with four of 22 (18.2%) classified as Low-DRS. All 22 simulation-only primary studies receive $C1 = 0.0$ (zero contribution from the 30%-weighted validation criterion), as no study includes hardware testbed or real-field validation. This distribution reflects the field’s structural simulation orientation rather than methodological deficiency, confirmed by the complementary MMAT assessment: six of 22 High (27.3%), 14 of 22 Moderate (63.6%), two of 22 Low (9.1%).

6.2. The Field Validation Gap: Structural Barriers and Pathways Forward

The predominance of simulation-only studies—22 of 22 primary research studies (100%); 22 of 29 included studies (75.9%)—is the most frequently cited limitation in this corpus. No primary study includes hardware testbed validation

or results from a real agricultural field environment. This complete absence of field-validated evidence reflects three structural barriers.

Economic barriers. Agricultural-grade UAV platforms capable of multi-agent coordination typically cost between USD 2,000 and 15,000 per unit [27]. A minimal swarm of five UAVs therefore requires USD 10,000–75,000 in hardware investment before accounting for field infrastructure, regulatory compliance, and personnel.

Regulatory barriers. Beyond-visual-line-of-sight (BVLOS) operations—required for autonomous multi-UAV coverage of large agricultural fields (>10 ha)—remain highly restricted in most jurisdictions.

Infrastructural barriers. Real agricultural field experiments require coordination between algorithm developers, agronomists, farm operators, and UAV pilots—a multidisciplinary infrastructure that most research groups lack.

6.3. *Environmental Modelling: The Wind Gap*

The absence of wind and weather modelling in 20 of 29 included studies (69.0%) represents a particularly acute gap for agricultural applications. For spraying applications, wind drift of pesticide or fertiliser droplets is both an efficacy issue and a regulatory compliance issue. The one spraying-focused primary study in this corpus (S13, Chu et al.) does not model this regulatory constraint, representing a complete absence of wind modelling in spraying-specific research.

6.4. *Metric Inconsistency: Implications for Algorithm Selection*

Over the full corpus of 29 studies, only 11 (37.9%) report execution time, four (13.8%) report energy consumption with explicit numerical values and units, and nine (31.0%) specify convergence criteria; over the 22 primary research studies, these rates are 11 of 22 (50.0%), four of 22 (18.2%), and nine of 22 (40.9%), respectively. No study reports all three simultaneously alongside field-relevant metrics such as coverage rate per hectare or mission completion probability under communication failures.

6.5. *Scalability and Heterogeneity*

Most reviewed studies evaluate algorithms on small UAV fleets of 2–5 agents in simulation environments covering less than 10 ha. Real precision agriculture operations routinely involve fields of 100–1,000+ ha where 10–50 UAVs may be required.

6.6. *Comparison with Adjacent Domains*

No primary study (zero of 22) includes any hardware component; no primary study reports full agricultural field validation (0% S+F). In search-and-rescue robotics, hardware validation rates in recent systematic reviews range from 40–60% [8]. The zero hardware validation rate in agricultural UAV SI research indicates the field has developed primarily as a sub-discipline of computational intelligence rather than as an applied robotics discipline.

6.7. Limitations of This Review

Several limitations must be acknowledged. First, the search was restricted to English-language publications in three databases, which may have excluded relevant work published in Chinese, Spanish, or other languages. Second, title and abstract screening was performed by the primary reviewer only. Third, the DRS instrument has not been validated as a psychometric tool. Fourth, the Gap 3 finding (15 of 29 studies with single-UAV experimental focus despite multi-UAV framing) may partially reflect a limitation of the inclusion criteria—the leniency in criterion I4b permitted entry of studies that validated multi-UAV frameworks with single agents, inflating the apparent frequency of this gap.

6.8. Implications for Agricultural Robotics Research Practice

Based on the gap analysis, we propose three concrete changes to standard research practice. First, all future SI-based agricultural UAV studies should adopt the **Staged Validation Protocol**: Simulation → Testbed → Controlled Field. Second, algorithm papers should report a minimum metric set of execution time, energy consumption, and convergence criteria alongside at least one agricultural-domain metric. Third, simulation environments should incorporate at minimum a simplified wind model and a temperature-dependent battery discharge curve.

6.9. Threats to Validity

Table 12 summarises the principal threats to the validity of this review and the mitigation strategies applied.

7. Research Agenda

7.1. Priority Directions for the Research Community

The gap analysis presented in Section 5 identifies 171 documented limitations across four dimensions, of which 76 (44.4%) directly prevent real-world deployment. The research agenda proposed here addresses the most fundamental of these through three synergistic priorities (Table 13). These priorities target the core clusters identified in Section 5.4 rather than providing a one-to-one mapping for the entire 171-gap taxonomy.

Priority 1 — Standardised Benchmark Framework (AgriSwarm-Bench). The absence of a shared evaluation framework is the foundational barrier that amplifies all other gaps [23]. AgriSwarm-Bench is designed to close this gap by providing the shared evaluation infrastructure the field currently lacks.

Priority 2 — Experimental Validation with Real UAVs. Absence of any hardware validation is the single most prevalent gap: no primary study (zero of 22, 0%) reports results from an operational agricultural field environment, and no primary study includes hardware testbed validation (zero of 22, 0%). When review articles are included, all 29 of 29 studies (100%) lack any hardware component. Closing this gap requires not only individual research groups conducting field experiments, but also a cultural shift toward treating simulation-only results as preliminary rather than definitive.

Priority 3 — Dynamic Environment Integration and Real-Time Replanning. Wind modelling, dynamic obstacle avoidance, and real-time path replanning are technically addressable with existing open-source tools but are absent from 25 of 29 reviewed studies (86.2%).

7.2. AgriSwarm-Bench: Technical Specification

Component 1 — Scenario Library. Eight canonical simulation scenarios covering principal agricultural mission types and field geometries:

- *S1–S2*: Cereal field coverage (100 ha and 500 ha), monitoring, wind speed 0 and 5 m/s.
- *S3–S4*: Orchard spraying (20 ha and 80 ha), row-based geometry, wind speed 3 m/s.
- *S5–S6*: Vineyard mapping (15 ha and 60 ha), narrow-corridor geometry.
- *S7–S8*: Heterogeneous field (200 ha), dynamic obstacles (farm machinery).

Component 2 — Unified Metric Suite. Six metrics required for leaderboard submission:

1. *Coverage rate (%)*: percentage of target area covered within mission time.
2. *Energy per hectare (Wh/ha)*: total fleet energy divided by area covered.
3. *Mission completion time (min)*: wall-clock time from takeoff to final landing.
4. *Scalability index*: ratio of coverage efficiency at ten UAVs to three UAVs.
5. *Convergence iterations*: SI algorithm iterations to reach within 5% of best-known optimum.
6. *Wind-drift deviation (m, spraying scenarios only)*: mean absolute lateral deviation under wind loading.

Component 3 — Open Dataset Repository. Community-contributed repository of real field experiment datasets following FAIR data principles, archived on Zenodo.

Component 4 — Community Leaderboard. Web-based leaderboard tracking algorithm performance across all 64 benchmark combinations, maintained at a public GitHub repository.

7.3. Proposed Community Roadmap

Table 14 presents a five-phase implementation roadmap.

The AgriSwarm-Bench framework is being piloted jointly at the Instituto Tecnológico de Chihuahua II (TecNM) and the Universidad Politécnica de Chihuahua (UPCh), Mexico.

8. Conclusions

8.1. Summary of Findings

This review analysed 29 peer-reviewed studies on swarm intelligence-based multi-UAV path planning in precision agriculture published between 2021 and 2025. Five principal findings emerge:

Table 12: Threats to validity and mitigation strategies.

Threat	Description	Mitigation
Publication bias	Positive results more likely published and indexed	Three independent databases; snowballing from seed papers
Search query limitations	Relevant papers using unlisted algorithm names may be missed	Query refined iteratively; sensitivity test with 5-paper gold-standard set (100% recall)
Single-database gap	Relevant papers in non-indexed venues not captured	MDPI manual search (zero eligible); OpenAlex supplementary search (125 records, zero new eligible); Scopus/WoS listed as Priority Future Work
Data extraction errors	Automated pipeline may misclassify fields	Automated extraction + manual verification for all 29 studies
Platform inclusion error	Non-UAV studies (ground robots) initially included	Four studies excluded post-hoc under criterion E6; corpus revised to n=29
Inter-rater variability	Single reviewer for T/A screening	Declared as limitation; second reviewer screens all 51 full-text records; $\kappa = 0.91$
Temporal bias (2025)	2025 data may be incomplete due to indexing lag	Explicitly noted; 2025 papers treated as preliminary indicator only

Table 13: Priority research directions based on gap analysis.

P	Contribution	Gaps	Impact	Key deliverable
1	Standardised benchmark framework (AgriSwarm-Bench)	18	High	Open repository with scenarios, metrics, leaderboard
2	Experimental validation: real UAVs in field	29	Very High	Staged Validation Protocol as community norm
3	Dynamic environment + real-time replanning	25	Very High	Wind/weather models in standard simulation pipelines

Table 14: Proposed community roadmap for AgriSwarm-Bench implementation. Phases 1–2 run in parallel during Year 1.

Ph.	Timeframe	Activity	Deliverable	Success metric
1	0–12 mo.	Implement AgriSwarm-Bench S1–S4 with PSO, ACO, ABC baselines	Open GitHub repository	≥ 3 algorithms benchmarked on S1–S4
2	0–12 mo.	Acquire 3–5 UAV platform; complete Stage 2 testbed validation	Hardware testbed	≥ 1 SI algorithm validated on physical UAVs
3	1–2 yr.	Real-time replanning module; add wind model to S5–S8	Enhanced algorithm	Wind-aware replanning demonstrated in S3–S4
4	1–2 yr.	Controlled field experiments; Stage 3 validation	Peer-reviewed paper	Field results for ≥ 2 scenarios with full metric suite
5	2–3 yr.	Release complete benchmark suite; community workshop	Community guidelines	≥ 5 external groups adopting AgriSwarm-Bench

- 1. PSO dominates the algorithm landscape** (nine of 29 studies, 31.0%; nine of 22 primary studies, 40.9%), followed by review articles (seven of 29, 24.1%), ACO variants (two of 29, 6.9%), and ABC (three of 29, 10.3%).
- 2. The field is overwhelmingly simulation-oriented:** all 22 of 22 primary research studies (100%) are simulation-only; no primary study includes hardware testbed validation (0% S+T); and no primary study reports results from a real agricultural field environment (0% S+F). When review articles are included, all 29 of 29 included studies (100%) lack any hardware component whatsoever. This is not primarily an algorithmic limitation but reflects structural barriers including platform cost (USD 2,000–15,000 per UAV), regulatory restrictions on BVLOS operations, and the absence of multidisciplinary research infrastructure.
- 3. Environmental factors are systematically neglected:** 25 of 29 studies (86.2%) do not model wind, weather, or dynamic obstacles. For the sole spraying-focused primary study (S13), wind and weather modelling is completely absent (1/1, 100%)—a particularly acute finding since wind drift affects both pesticide efficacy and regulatory compliance.
- 4. Metric reporting is fragmented:** over the full corpus, only 11 of 29 (37.9%) report execution time, four of 29 (13.8%) quantitative energy consumption, and nine of 29 (31.0%) convergence criteria; over the 22 primary studies, these rates are 11 of 22 (50.0%), four of 22 (18.2%), and nine of 22 (40.9%), respectively.
- 5. 171 research gaps were documented** across four dimensions—Technological (51 of 171, 29.8%), Practical (52 of 171, 30.4%), Methodological (39 of 171, 22.8%), and Theoretical (29 of 171, 17.0%)—of which 76 (44.4%) are classified as Critical barriers to real-world deployment.

8.2. Broader Implications

For the computational intelligence community: The 100% simulation-only rate among primary studies (22 of 22), with zero studies achieving any hardware validation, and the fragmented metric landscape indicate that the incentive structure of the field is producing a literature with limited deployability. AgriSwarm-Bench offers one mechanism to shift this incentive structure.

For the precision agriculture community: The review demonstrates that SI algorithms for multi-UAV coordination have reached algorithmic maturity sufficient to address agricultural-scale tasks, but the translation to deployment requires investment in regulatory frameworks, field testing infrastructure, and domain-specific benchmarks.

For journal editors and funding agencies: The DRS results (four of 22 primary studies classified as Low-DRS, 18.2%; four as High-DRS, 18.2%) indicate that current peer review standards are not adequately filtering for deployment relevance. Adopting the Staged Validation Protocol and the minimum metric reporting standard as explicit acceptance criteria would accelerate the field’s transition to deployable agricultural technology.

8.3. Limitations

The principal limitations are: (i) restriction to three English-language databases with no access to Scopus or Web of Science; (ii) single-reviewer title/abstract screening; (iii) absence of prospective protocol registration (mitigated by retrospective OSF registration, <https://doi.org/10.17605/OSF.IO/64DQ9>); (iv) DRS instrument not independently validated for psychometric properties; and (v) four studies excluded post-hoc under criterion E6 (non-UAV platforms), reducing the corpus from 33 to 29 studies.

8.4. Future Work

1. **Extend the search** to Scopus and Web of Science, and include non-English publications, to assess whether the algorithm distribution and gap patterns are robust to a broader search.
2. **Implement AgriSwarm-Bench Phases 1–2** at Instituto Tecnológico de Chihuahua II: develop the S1–S4 benchmark scenarios, establish PSO/ACO/ABC baselines, and complete Stage 2 hardware testbed validation with a three-to-five UAV platform.
3. **Release a public dataset** from the first controlled field experiment on Mendeley Data.
4. **Conduct a living review update** following the AgriSwarm-Bench roadmap milestones: re-screen Scopus, WoS, and CNKI annually and update the gap taxonomy with each new cohort of studies.

Declaration of Competing Interest

The authors declare no financial conflicts of interest. Two potential non-financial conflicts of interest are disclosed. First, the corresponding author’s institutions (Instituto Tecnológico de Chihuahua II and Universidad Politécnica de Chihuahua) are proposed as pilot sites for the AgriSwarm-Bench implementation described in Section 7. Second,

the second reviewer (Dr. Hernán de la Garza Gutiérrez, co-author) also serves as doctoral thesis supervisor of the corresponding author (A. Hidalgo Morales), which may constitute a non-financial conflict of interest in the inter-rater screening process; mitigated by the structured discussion protocol and near-perfect independent agreement ($\kappa = 0.91$).

Data Availability Statement

The review protocol, search strategies, extraction tools, and supplementary materials are publicly available at the Open Science Framework: <https://doi.org/10.17605/OSF.IO/64DQ9>. The master extraction database, Python scripts, and publication-ready figures will be deposited in Mendeley Data upon manuscript acceptance.

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Hernán de la Garza Gutiérrez: Supervision, Methodology, Validation (independent study selection, inter-rater agreement), Formal Analysis (Cohen’s kappa calculation), Writing – Review & Editing.

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524 **List of Abbreviations**

525 **ABC** Artificial Bee Colony

526 **ACO** Ant Colony Optimisation

527 **ANFIS** Adaptive Neuro-Fuzzy Inference System

528 **APF** Artificial Potential Field

529 **BVLOS** Beyond Visual Line of Sight

530 **CI** Confidence Interval

531 **CRedit** Contributor Roles Taxonomy

532 **DE** Differential Evolution

533 **DOI** Digital Object Identifier

534 **DRS** Deployment Readiness Score

535 **FAA** Federal Aviation Administration (USA)

536 **GA** Genetic Algorithm

537 **GCS** Ground Control Station

538 **GNSS** Global Navigation Satellite System

539 **GPS** Global Positioning System

540 **GWO** Grey Wolf Optimiser

541 **ICAO** International Civil Aviation Organization

542 **IEEE** Institute of Electrical and Electronics Engineers

543 **IMU** Inertial Measurement Unit

544 **LAI** Leaf Area Index

545 **LiDAR** Light Detection and Ranging

546 **MMAT** Mixed Methods Appraisal Tool

547 **MPC** Model Predictive Control

548 **NDVI** Normalized Difference Vegetation Index

549 **OSF** Open Science Framework

550 **PA** Precision Agriculture

551 **PICOC** Population, Intervention, Comparison, Outcome, Context

552 **PRISMA** Preferred Reporting Items for Systematic Reviews and Meta-Analyses

553 **PSO** Particle Swarm Optimisation

554 **RPAS** Remotely Piloted Aircraft System (ICAO official term)

555 **RQ** Research Question

556 **SI** Swarm Intelligence

557 **SLAM** Simultaneous Localization and Mapping

558 **UAS** Unmanned Aircraft System (FAA term; equivalent to RPAS)

559 **UAV** Unmanned Aerial Vehicle (generic technical term)

560 **VRA** Variable Rate Application

561 **WoS** Web of Science

562 **References**

- 563 [1] C. Zhang, J. M. Kovacs, The application of small unmanned aerial systems for precision agriculture, *Precision*
564 *Agriculture* 13 (4) (2012) 393–408. doi:10.1007/s11119-012-9274-5.
- 565 [2] D. C. Tsouros, S. Bibi, P. G. Sarigiannidis, A review on UAV-based applications for precision agriculture, *Infor-*
566 *mation* 10 (11) (2019) 349. doi:10.3390/info10110349.
- 567 [3] V. Puri, A. Nayyar, L. Raja, Agriculture drones: A modern breakthrough in precision agriculture, *Journal of*
568 *Statistics and Management Systems* 20 (4) (2017) 507–518. doi:10.1080/09720510.2017.1395171.
- 569 [4] U. R. Mogili, B. B. V. L. Deepak, Review on application of drone systems in precision agriculture, *Procedia*
570 *Computer Science* 133 (2018) 502–509. doi:10.1016/j.procs.2018.07.063.
- 571 [5] E. R. Hunt, W. D. Hively, C. S. T. Daughtry, Evaluation of UAV imagery for agricultural crop monitoring, *Remote*
572 *Sensing* 10 (4) (2018) 595. doi:10.3390/rs10040595.
- 573 [6] Grand View Research, Uav market size, share & trends analysis report, accessed: 2026-04-07 (2024).

- [7] MarketsandMarkets, Unmanned aerial vehicle (uav) market by platform, type, application, and region - global forecast to 2028, accessed: 2026-04-07 (2023).
- [8] S. J. Chung, A. A. Paranjape, P. Dames, S. Shen, V. Kumar, A survey on aerial swarm robotics, *IEEE Transactions on Robotics* 34 (4) (2018) 837–855. doi:10.1109/TR0.2018.2857475.
- [9] L. Gupta, R. Jain, G. Vaszkun, Survey of UAV communication networks, *IEEE Communications Surveys and Tutorials* 18 (4) (2016) 2595–2621. doi:10.1109/COMST.2015.2495297.
- [10] J. Huang, Y. Luo, Q. Quan, B. Wang, X. Xue, Y. Zhang, An autonomous task assignment and decision-making method for coverage path planning of multiple pesticide spraying UAVs, *Computers and Electronics in Agriculture* 212 (2023) 108128. doi:10.1016/j.compag.2023.108128.
- [11] I. Bekmezci, O. K. Sahingoz, S. Temel, Flying ad-hoc networks (FANETs): A survey, *Ad Hoc Networks* 11 (3) (2013) 1254–1270. doi:10.1016/j.adhoc.2012.12.004.
- [12] S. Hayat, E. Yanmaz, C. Bettstetter, Experimental analysis of multipoint-to-point UAV communications, *IEEE Communications Letters* 20 (7) (2016) 1433–1436. doi:10.1109/LCOMM.2016.2561274.
- [13] A. Ait Saadi, A. Soukane, Y. Meraihi, A. B. Gabis, S. Mirjalili, A. Ramdane-Cherif, UAV path planning using optimization approaches: A survey, *Archives of Computational Methods in Engineering* 29 (6) (2022) 4233–4284. doi:10.1007/s11831-022-09742-7.
- [14] M. Dorigo, M. Birattari, T. Stutzle, Ant colony optimization, *IEEE Computational Intelligence Magazine* 1 (4) (2006) 28–39. doi:10.1109/MCI.2006.329691.
- [15] J. Kennedy, R. Eberhart, Particle swarm optimization, in: *Proceedings of ICNN'95 – International Conference on Neural Networks*, Vol. 4, 1995, pp. 1942–1948. doi:10.1109/ICNN.1995.488968.
- [16] C. Blum, X. Li, Swarm intelligence in optimization, in: *Swarm Intelligence: Introduction and Applications*, Springer, 2008, pp. 43–85. doi:10.1007/978-3-540-74089-6_2.
- [17] X. S. Yang, *Engineering Optimization: An Introduction with Metaheuristic Applications*, John Wiley & Sons, 2010. doi:10.1002/9780470640425.
- [18] G. Beni, From swarm intelligence to swarm robotics, in: *Swarm Robotics*, Springer, 2005, pp. 1–9. doi:10.1007/978-3-540-30552-1_1.
- [19] E. Bonabeau, M. Dorigo, G. Theraulaz, *Swarm Intelligence: From Natural to Artificial Systems*, Oxford University Press, New York, 1999, ISBN: 978-0-19-513159-8. No DOI assigned (pre-digital monograph, 1999).

- [20] X. S. Yang, M. Karamanoglu, X. He, Flower pollination algorithm: A novel approach for multiobjective optimization, *Engineering Optimization* 46 (9) (2014) 1222–1237, published online 2013; formal print volume year 2014. Cite key yang2013 retained for tex compatibility. doi:10.1080/0305215X.2013.832237.
- [21] C. Xiao, H. Yang, B. Zhang, Multi-unmanned aerial vehicle path planning based on improved nutcracker optimization algorithm, *Drones* 9 (1) (2025) 38. doi:10.3390/drones9010038.
- [22] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M. M. Lalu, T. Li, E. W. Loder, E. Mayo-Wilson, S. McDonald, L. A. McGuinness, L. A. Stewart, J. Thomas, A. C. Tricco, V. A. Welch, P. Whiting, D. Moher, The PRISMA 2020 statement: An updated guideline for reporting systematic reviews, *BMJ* 372 (2021) n71. doi:10.1136/bmj.n71.
- [23] M. Brambilla, E. Ferrante, M. Birattari, M. Dorigo, Swarm robotics: A review from the swarm engineering perspective, *Swarm Intelligence* 7 (1) (2013) 1–41. doi:10.1007/s11721-012-0075-2.
- [24] L. Bayindir, A review of swarm robotics tasks, *Neurocomputing* 172 (2016) 292–321. doi:10.1016/j.neucom.2015.05.116.
- [25] A. Puente-Castro, D. Rivero, A. Pazos, E. Fernandez-Blanco, A review of artificial intelligence applied to path planning in UAV swarms, *Neural Computing and Applications* 34 (1) (2022) 153–170. doi:10.1007/s00521-021-06569-4.
- [26] A. Sharma, S. Shoval, A. Sharma, J. K. Pandey, Path planning for multiple targets interception by the swarm of UAVs based on swarm intelligence algorithms: A review, *IETE Technical Review* 39 (3) (2022) 524–538. doi:10.1080/02564602.2021.1894250.
- [27] M. D. Phung, Q. P. Ha, Safety-enhanced UAV path planning with spherical vector-based particle swarm optimization, *Applied Soft Computing* 107 (2021) 107376. doi:10.1016/j.asoc.2021.107376.
- [28] M. Shafiq, Z. A. Ali, A. Israr, E. H. Alkhamash, M. Hadjouni, J. J. Jussila, Convergence analysis of path planning of multi-UAVs using max-min ant colony optimization approach, *Sensors* 22 (14) (2022) 5395. doi:10.3390/s22145395.
- [29] Q. Liu, Y. Zhang, M. Li, Z. Zhang, N. Cao, J. Shang, Multi-UAV path planning based on fusion of sparrow search algorithm and improved bioinspired neural network, *IEEE Access* 9 (2021) 120946–120960. doi:10.1109/ACCESS.2021.3109879.
- [30] N. Pan, M. Zhang, Y. Sun, S. Chen, H. Liu, X. Guo, Study on border patrol task planning of heterogeneous UAVs group based on swarm intelligence, *Science Progress* 105 (4) (2022) 00368504221094722, *Science Progress* uses article IDs in place of page ranges; ID matches DOI suffix. doi:10.1177/00368504221094722.

- [31] Q. N. Hong, S. Fàbregues, G. Bartlett, F. Boardman, M. Cargo, P. Dagenais, M.-P. Gagnon, F. Griffiths, B. Nicolau, A. O’Cathain, P. Pluye, The mixed methods appraisal tool (MMAT) version 2018 for information professionals and researchers, *Education for Information* 34 (4) (2018) 285–291. doi:10.3233/EFI-180221.
- [32] J. Chen, F. Ling, Y. Zhang, T. You, Y. Liu, X. Du, Coverage path planning of heterogeneous unmanned aerial vehicles based on ant colony system, *Swarm and Evolutionary Computation* 65 (2021) 101005. doi:10.1016/j.swevo.2021.101005.
- [33] S. Lin, F. Li, X. Li, K. Jia, X. Zhang, Improved artificial bee colony algorithm based on multi-strategy synthesis for UAV path planning, *IEEE Access* 10 (2022) 119269–119285. doi:10.1109/ACCESS.2022.3218685.
- [34] G. Liu, C. Shu, Z. Liang, B. Peng, L. Cheng, A modified sparrow search algorithm with application in 3D route planning for UAV, *Sensors* 21 (4) (2021) 1224. doi:10.3390/s21041224.
- [35] H. Liu, J. Ge, Y. Wang, J. Li, K. Ding, Z. Zhang, Z. Guo, W. Li, J. Lan, Multi-UAV optimal mission assignment and path planning for disaster rescue using adaptive genetic algorithm and improved artificial bee colony method, *Actuators* 11 (1) (2021) 4. doi:10.3390/act11010004.
- [36] A. T. Mathew, A. Paul, A. Rojan, A. Thomas, Implementation of swarm intelligence algorithms for path planning, *Journal of Physics: Conference Series* 1831 (1) (2021) 012008. doi:10.1088/1742-6596/1831/1/012008.
- [37] C. Ntakolia, D. V. Lyridis, A swarm intelligence graph-based pathfinding algorithm based on fuzzy logic (SIG-PAF), *Journal of Marine Science and Engineering* 9 (11) (2021) 1243. doi:10.3390/jmse9111243.
- [38] W. Yu, J. Liu, J. Zhou, A novel sparrow particle swarm algorithm (SPSA) for unmanned aerial vehicle path planning, *Scientific Programming* 2021 (2021) 5158304. doi:10.1155/2021/5158304.
- [39] H. Chu, J. Yi, F. Yang, Chaos particle swarm optimization enhancement algorithm for UAV safe path planning, *Applied Sciences* 12 (18) (2022) 8977. doi:10.3390/app12188977.
- [40] G. Fevgas, T. Lagkas, V. Argyriou, P. Sarigiannidis, Coverage path planning methods focusing on energy efficient and cooperative strategies for unmanned aerial vehicles, *Sensors* 22 (3) (2022) 1235. doi:10.3390/s22031235.
- [41] A. Israr, Z. A. Ali, E. H. Alkhamash, J. J. Jussila, Optimization methods applied to motion planning of unmanned aerial vehicles: A review, *Drones* 6 (5) (2022) 126. doi:10.3390/drones6050126.
- [42] Y. Ji, X. Zhao, J. Hao, A novel UAV path planning algorithm based on double-dynamic biogeography-based learning particle swarm optimization, *Mobile Information Systems* 2022 (2022) 8519708. doi:10.1155/2022/8519708.

- [43] B. Selma, S. Chouraqui, B. Selma, H. Abouaissa, Design an optimal ANFIS controller using bee colony optimization for trajectory tracking of a quadrotor UAV, *Journal of The Institution of Engineers (India): Series B* 103 (5) (2022) 1505–1519. doi:10.1007/s40031-022-00747-1.
- [44] N. Ahmed, C. J. Pawase, K. Chang, Distributed 3-D path planning for multi-UAVs with full area surveillance based on particle swarm optimization, *Applied Sciences* 11 (8) (2021) 3417. doi:10.3390/app11083417.
- [45] S. Xu, L. Li, Z. Zhou, Y. Mao, J. Huang, A task allocation strategy of the UAV swarm based on multi-discrete wolf pack algorithm, *Applied Sciences* 12 (3) (2022) 1331. doi:10.3390/app12031331.
- [46] L. Deng, H. Chen, X. Zhang, H. Liu, Three-dimensional path planning of UAV based on improved particle swarm optimization, *Drones* 7 (4) (2023) 243. doi:10.3390/drones7040243.
- [47] C. Rao, Z. Wang, P. Shao, A multi-strategy collaborative grey wolf optimization algorithm for UAV path planning, *Drones* 8 (2) (2024) 47. doi:10.3390/drones8020047.
- [48] W. Zhang, W. Zhang, An efficient UAV localization technique based on particle swarm optimization, *IEEE Transactions on Vehicular Technology* 71 (9) (2022) 9990–10002. doi:10.1109/TVT.2022.3178228.
- [49] L. Zuo, Y. Wang, Y. Lu, R. Gu, Hybrid APF–PSO algorithm for regional dynamic formation of UAV swarms, *Drones* 9 (2) (2025) 112. doi:10.3390/drones9020112.
- [50] X. Yang, S. Zhao, W. Gao, P. Li, Z. Feng, L. Li, T. Jia, X. Wang, Three-dimensional path planning for UAV based on multi-strategy dream optimization algorithm, *Drones* 9 (1) (2025) 55. doi:10.3390/drones9010055.
- [51] Y. Yang, X. Xiong, Y. Yan, UAV formation trajectory planning algorithms: A review, *Drones* 7 (1) (2023) 62. doi:10.3390/drones7010062.
- [52] J. Tang, H. Duan, S. Lao, Swarm intelligence algorithms for multiple unmanned aerial vehicles collaboration: A comprehensive review, *Artificial Intelligence Review* 56 (5) (2022) 4295–4327. doi:10.1007/s10462-022-10281-7.
- [53] International Organization for Standardization, Space systems — space product assurance — technology readiness levels (TRL), International Standard ISO 16290:2013, ISO, defines TRL 1–9 scale for technology maturity assessment (2013).
- [54] R. Olfati-Saber, Flocking for multi-agent dynamic systems: Algorithms and theory, *IEEE Transactions on Automatic Control* 51 (3) (2006) 401–420. doi:10.1109/TAC.2005.864190.
- [55] L. Meier, D. Honegger, M. Pollefeys, PX4: A node-based multithreaded open source robotics framework for deeply embedded platforms, in: *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, Seattle, WA, USA, 2015, pp. 6235–6240. doi:10.1109/ICRA.2015.7140074.